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1  -- Connecting to our HealthCapstone database
2  USE HealthCapstone
3
4  -- Counting the number of rows in our newly imported data
5  SELECT COUNT(*) AS total_rows
6  FROM insurance
7
8  -- What is the average insurance charge for smokers vs non-smokers?
9  SELECT smoker, AVG(charges) AS average_charge
10 FROM insurance
11 GROUP BY smoker
12
13 -- Which age groups have the highest average insurance cost
14 SELECT (
15     CASE
16         WHEN age BETWEEN 18 AND 29 THEN '18-29'
17         WHEN age BETWEEN 30 AND 39 THEN '30-39'
18         WHEN age BETWEEN 40 AND 49 THEN '40-49'
19         WHEN age BETWEEN 50 AND 59 THEN '50-59'
20         ELSE '60+'
21     END
22 ) AS age_group,
23     AVG(charges) AS average_charge
24 FROM insurance
25 GROUP BY (
26     CASE
27         WHEN age BETWEEN 18 AND 29 THEN '18-29'
28         WHEN age BETWEEN 30 AND 39 THEN '30-39'
29         WHEN age BETWEEN 40 AND 49 THEN '40-49'
30         WHEN age BETWEEN 50 AND 59 THEN '50-59'
31         ELSE '60+'
32     END
33 )
34 ORDER BY average_charge DESC
35
36 -- How does BMI category affect average charges?
37 SELECT (
38     CASE
39         WHEN bmi < 18.5 THEN 'Underweight'
40         WHEN bmi >= 18.5 AND bmi < 25 THEN 'Normal weight'
41         WHEN bmi >= 25 AND bmi < 30 THEN 'Overweight'
42         ELSE 'Obesity'
43     END
44 ) AS bmi_category,
45     AVG(charges) AS average_charge
46 FROM insurance
47 GROUP BY (
48     CASE
49         WHEN bmi < 18.5 THEN 'Underweight'
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50         WHEN bmi >= 18.5 AND bmi < 25 THEN 'Normal weight'
51         WHEN bmi >= 25 AND bmi < 30 THEN 'Overweight'
52         ELSE 'Obesity'
53     END
54 )
55 ORDER BY average_charge DESC
56
57 /* Extra
58 What is the average insurance charge for each region, from highest to
    lowest? */
59 SELECT region, AVG(charges) AS average_charge
60 FROM insurance
61 GROUP BY region
62 ORDER BY average_charge DESC
```